

MVLU COLLEGE

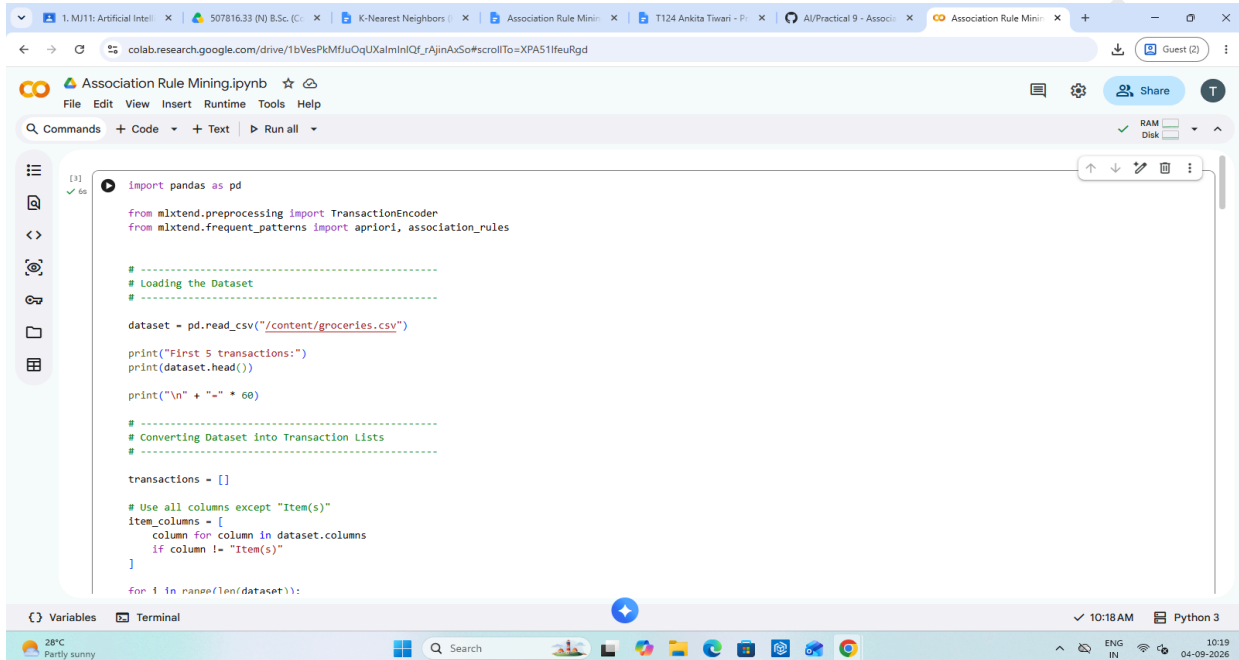
AIM: Association Rule Mining

Implement the Association Rule Mining algorithm (e.g., Apriori) to find frequent itemsets.

Generate association rules from the frequent itemsets and calculate their support and confidence.

Interpret and analyze the discovered association rules.

INPUT:



```
[3]: import pandas as pd
from mlxtend.preprocessing import TransactionEncoder
from mlxtend.frequent_patterns import apriori, association_rules

# -----
# Loading the Dataset
# -----

dataset = pd.read_csv("/content/groceries.csv")

print("First 5 transactions:")
print(dataset.head())

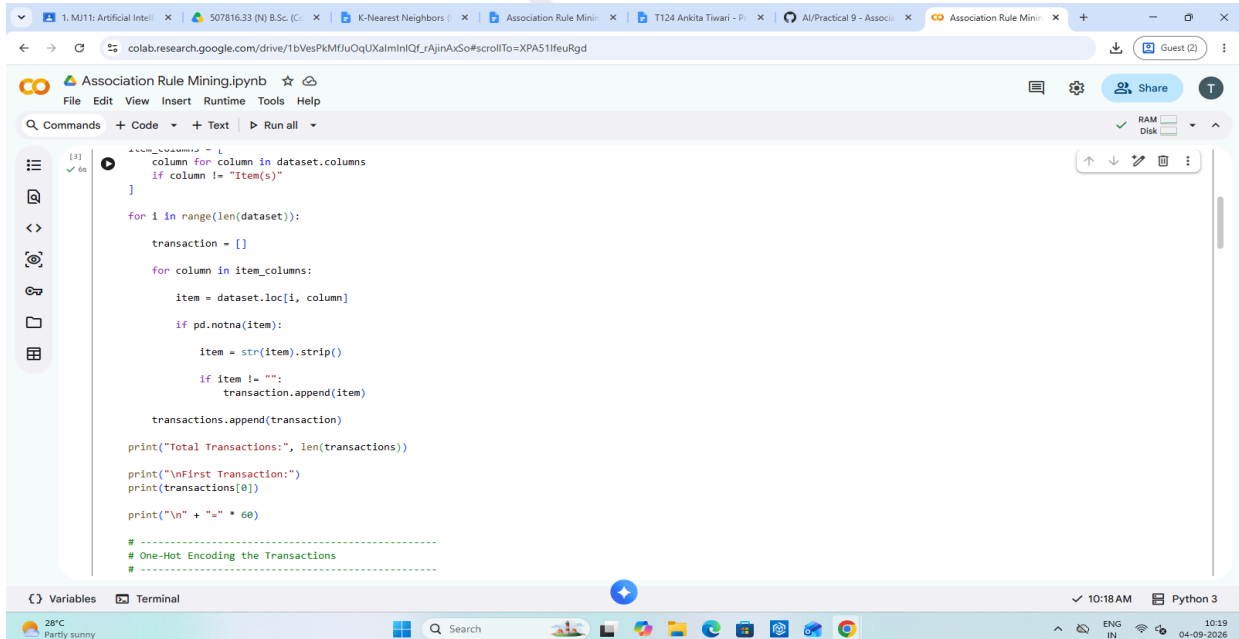
print("\n" + "-" * 60)

# -----
# Converting Dataset into Transaction Lists
# -----

transactions = []

# Use all columns except "Item(s)"
item_columns = [
    column for column in dataset.columns
    if column != "Item(s)"
]

for i in range(len(dataset)):
```



```
        column for column in dataset.columns
        if column != "Item(s)"
    ]

    for i in range(len(dataset)):

        transaction = []

        for column in item_columns:

            item = dataset.loc[i, column]

            if pd.notna(item):

                item = str(item).strip()

                if item != "":
                    transaction.append(item)

        transactions.append(transaction)

    print("Total Transactions:", len(transactions))

    print("\nFirst Transaction:")
    print(transactions[0])

    print("\n" + "-" * 60)

# -----
# One-Hot Encoding the Transactions
# -----
```

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```
encoder = TransactionEncoder()

encoded_data = encoder.fit(
    transactions
).transform(
    transactions
)

basket = pd.DataFrame(
    encoded_data,
    columns=encoder.columns_
)

print("Transaction Data after Encoding:")
print(basket.head())

print("\n" + "=" * 60)

# -----
# Finding Frequent Itemsets using Apriori
# -----

frequent_itemsets = apriori(
    basket,
    min_support=0.01,
    use_colnames=True
)

# Sorting frequent itemsets by support
frequent_itemsets = frequent_itemsets.sort_values(
    by="support"
)
```

```
use_colnames=True
)

# Sorting frequent itemsets by support
frequent_itemsets = frequent_itemsets.sort_values(
    by="support",
    ascending=False
)

print("Frequent Itemsets:")
print(frequent_itemsets.head(10))

print("\n" + "=" * 60)

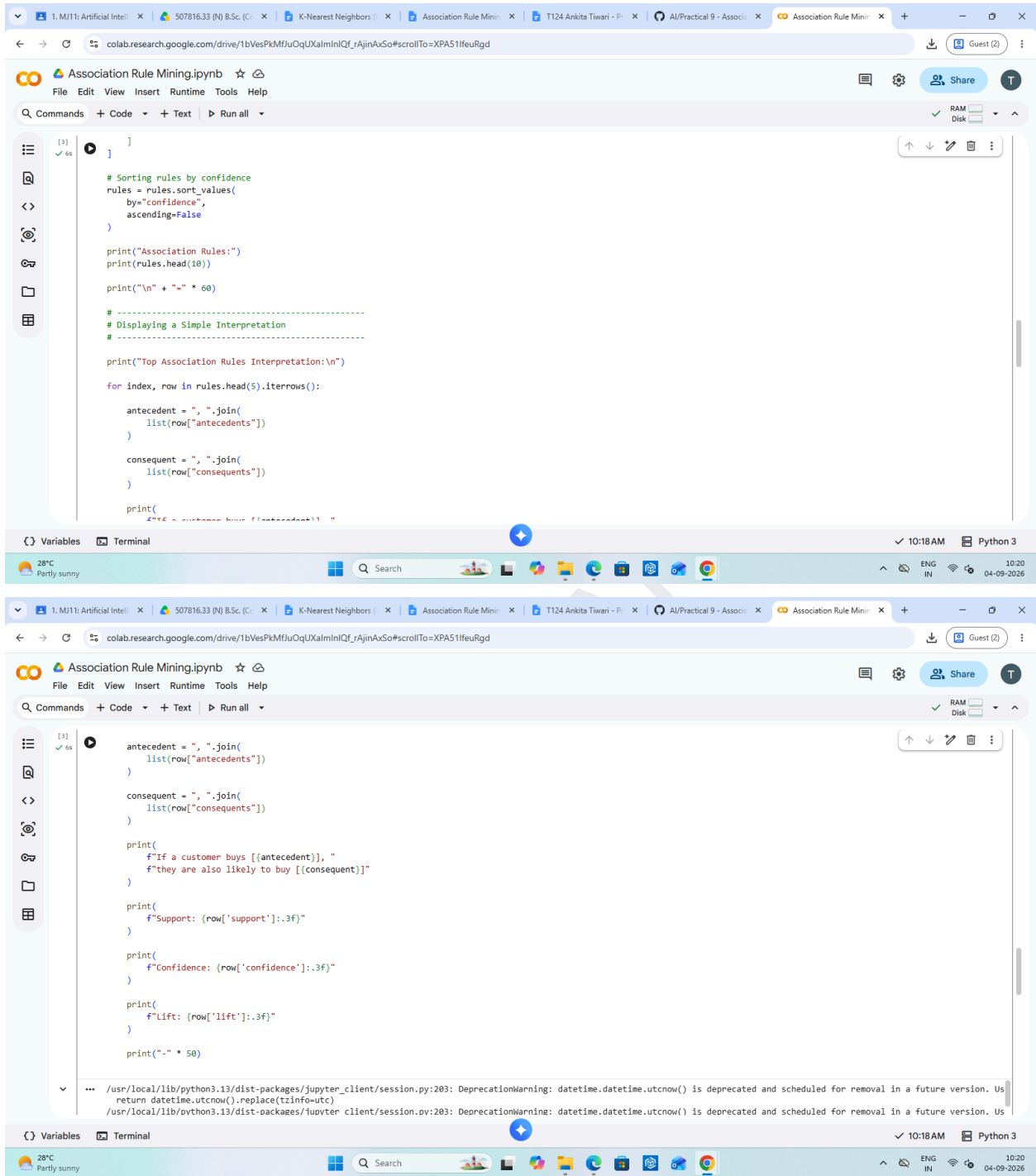
# -----
# Generating Association Rules
# -----

rules = association_rules(
    frequent_itemsets,
    metric="confidence",
    min_threshold=0.30
)

# Keeping important columns
rules = rules[
    [
        "antecedents",
        "consequents",
        "support",
        "confidence",
        "lift"
    ]
]
```

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```
[3] ✓ ds
]

# Sorting rules by confidence
rules = rules.sort_values(
    by="confidence",
    ascending=False
)

print("Association Rules:")
print(rules.head(10))

print("\n" + "-" * 60)

# -----
# Displaying a Simple Interpretation
# -----

print("Top Association Rules Interpretation:\n")

for index, row in rules.head(5).iterrows():

    antecedent = ", ".join(
        list(row["antecedents"])
    )

    consequent = ", ".join(
        list(row["consequents"])
    )

    print(
        f"If a customer buys {antecedent}, "
        f"they are also likely to buy {consequent}"
    )

    print(
        f"Support: {row['support']:.3f}"
    )

    print(
        f"Confidence: {row['confidence']:.3f}"
    )

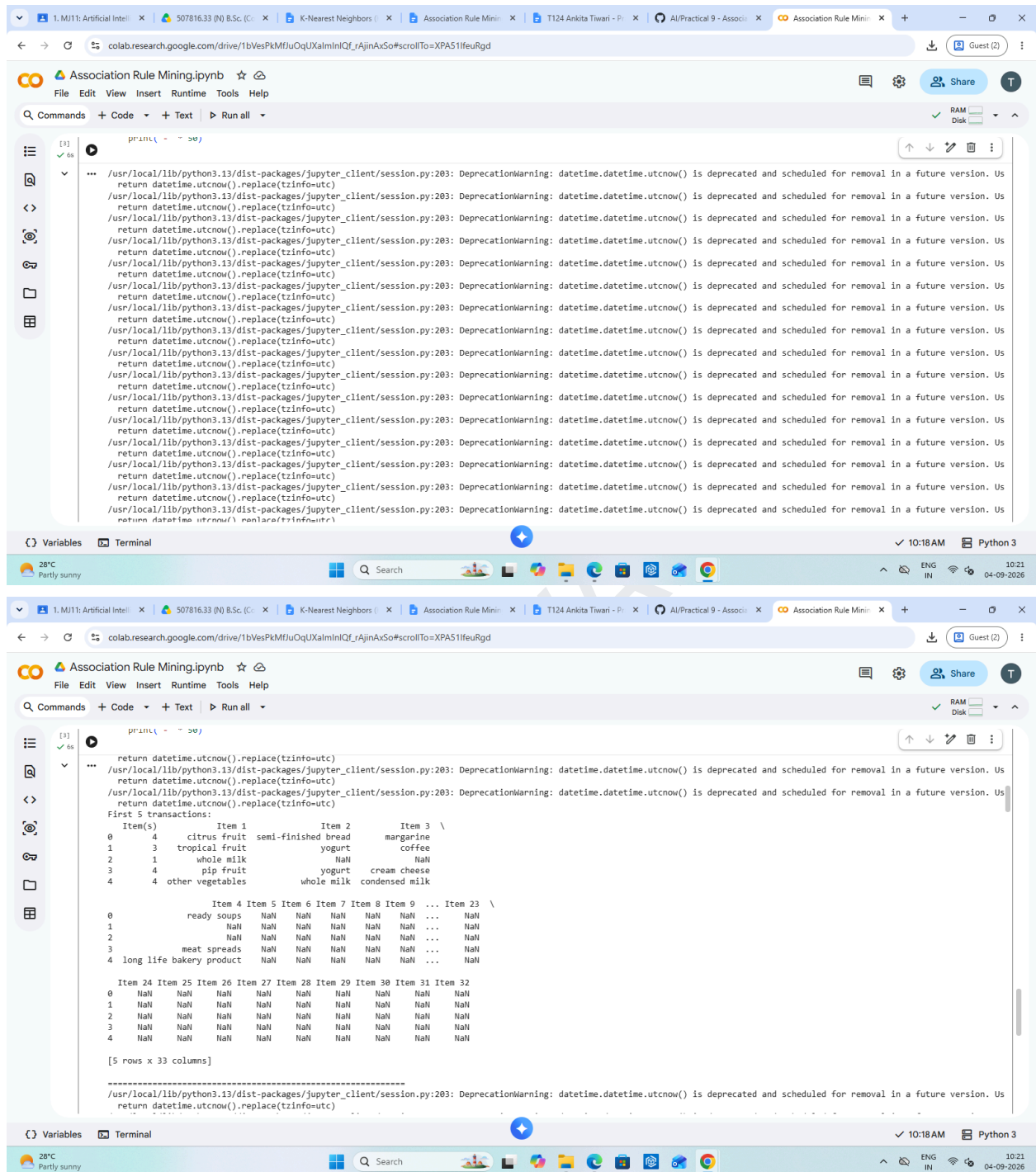
    print(
        f"Lift: {row['lift']:.3f}"
    )

    print("-" * 50)

... /usr/local/lib/python3.13/dist-packages/jupyter_client/session.py:203: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Us
return datetime.utcnow().replace(tzinfo=utc)
/usr/local/lib/python3.13/dist-packages/jupyter_client/session.py:203: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Us
```

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OUTPUT:



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The image displays two screenshots of a Google Colab notebook titled "Association Rule Mining.ipynb". The notebook is running on a virtual machine with Python 3. The first screenshot shows the output of a transaction data encoding process, displaying a matrix of boolean values (True/False) for various items across five transactions. The second screenshot shows the output of an association rule mining process, displaying frequent itemsets and association rules with their support, confidence, and lift values.

Transaction Data after Encoding:

	Instant food products	UHT-milk	abrasive cleaner	artif. sweetener	
0	False	False	False	False	
1	False	False	False	False	
2	False	False	False	False	
3	False	False	False	False	
4	False	False	False	False	

	baby cosmetics	baby food	bags	baking powder	bathroom cleaner	beef	
0	False	False	False	False	False	False	
1	False	False	False	False	False	False	
2	False	False	False	False	False	False	
3	False	False	False	False	False	False	
4	False	False	False	False	False	False	

	turkey	vinegar	waffles	whipped/sour cream	whisky	white bread	
0	False	False	False	False	False	False	
1	False	False	False	False	False	False	
2	False	False	False	False	False	False	
3	False	False	False	False	False	False	
4	False	False	False	False	False	False	

	white wine	whole milk	yogurt	zwieback	
0	False	False	False	False	
1	False	False	True	False	
2	False	True	False	False	
3	False	False	True	False	
4	False	True	False	False	

[5 rows x 169 columns]

Frequent Itemsets:

support	itemsets
86 0.255516	(whole milk)
55 0.193493	(other vegetables)
66 0.183935	(rolls/buns)
75 0.174377	(soda)
87 0.139582	(yogurt)
6 0.118524	(bottled water)
67 0.108998	(root vegetables)
81 0.104931	(tropical fruit)
73 0.098526	(shopping bags)
70 0.093950	(sausage)

Association Rules:

antecedents	consequents	support	
(citrus fruit, root vegetables)	(other vegetables)	0.018371	
(tropical fruit, root vegetables)	(other vegetables)	0.012383	
(yogurt, curd)	(whole milk)	0.010066	
(butter, other vegetables)	(whole milk)	0.011490	
(tropical fruit, root vegetables)	(whole milk)	0.011998	
(yogurt, root vegetables)	(whole milk)	0.014540	
(domestic eggs, other vegetables)	(whole milk)	0.012383	
(yogurt, whipped/sour cream)	(whole milk)	0.010880	
(rolls/buns, root vegetables)	(whole milk)	0.012710	
(other vegetables, pip fruit)	(whole milk)	0.013523	

confidence	lift
112 0.586207	3.029608
83 0.584541	3.028999
122 0.582353	2.279125

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The image displays two screenshots of a Jupyter Notebook interface, likely Google Colab, showing the results of an Association Rule Mining (ARM) analysis. The notebook is titled "Association Rule Mining.ipynb".

Top Screenshot:

- The code cell shows a table of itemsets and their associated metrics (Support, Confidence, Lift).
- The output displays the top association rules and their interpretation.

Itemset	Support	Confidence	Lift
(yogurt, whipped/sour cream)	0.010880	0.010880	0.010880
(rolls/buns, root vegetables)	0.012710	0.012710	0.012710
(other vegetables, pip fruit)	0.013523	0.013523	0.013523

Top Association Rules Interpretation:

- If a customer buys [citrus fruit, root vegetables], they are also likely to buy [other vegetables]
Support: 0.010
Confidence: 0.586
Lift: 3.030
- If a customer buys [tropical fruit, root vegetables], they are also likely to buy [other vegetables]
Support: 0.012
Confidence: 0.585
Lift: 3.021
- If a customer buys [yogurt, curd], they are also likely to buy [whole milk]
Support: 0.010
Confidence: 0.582
Lift: 2.279

Bottom Screenshot:

- The code cell shows the same ARM analysis, but the output displays a different set of rules.
- The output also includes a warning message about a deprecated datetime function.

Warning Message:

```
/usr/local/lib/python3.13/dist-packages/jupyter_client/session.py:203: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Use datetime.datetime.now(datetime.UTC).replace(tzinfo=utc) instead.
```

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